MODULE – 2

LITERATURE REVIEW AND TECHNICAL READING/ ATTRIBUTIONS AND CITATIONS

Literature Review and Technical Reading:

Introduction:

A literature review is a survey of scholarly sources on a specific topic. It provides an overview of current knowledge, allowing you to identify relevant theories, methods, and gaps in the existing research that you can later apply to your paper, thesis, or dissertation topic.

There are five key steps to writing a literature review:

1. **Search** for relevant literature
2. **Evaluate** sources
3. **Identify** themes, debates, and gaps
4. **Outline** the structure
5. **Write** your literature review

What is the purpose of a literature review?

The reasons for review of related literature are:

- To gain a background knowledge of the research topic.
- To identify the concepts relating to it, potential relationships between them and to formulate researchable hypotheses.
- To identify appropriate methodology, research design, methods, of measuring concepts and techniques of analysis.
- To identify data sources used by other researchers and To learn how others structured their reports.

When you write a thesis, dissertation, or research paper, you will likely have to conduct a literature review to situate your research within existing knowledge. The literature review gives you a chance to:

- Demonstrate your familiarity with the topic and its scholarly context
- Develop a theoretical framework and methodology for your research
- Position your work in relation to other researchers and theorists
- Show how your research addresses a gap or contributes to a debate
- Evaluate the current state of research and demonstrate your knowledge of the scholarly debates around your topic.

Writing literature reviews is a particularly important skill if you want to apply for graduate school or pursue a career in research.

New and Existing Knowledge:

In the vast landscape of research and professional development, literature reviews stand as critical landmarks, illuminating the path of understanding and discovery. Whether you're an academic researcher, a corporate strategist, or a curious learner, conducting a literature review can be a transformative experience. Here's a step-by-step guide to help you navigate this essential process according to best practices.

Step 1 – Search for relevant literature

Before you begin searching for literature, you need a clearly defined topic. If you are writing the literature review section of a dissertation or research paper, you will search for literature related to your research problem and questions.

Make a list of keywords

Start by creating a list of keywords related to your research question. Include each of the key concepts or variables you're interested in, and list any synonyms and related terms. You can add to this list as you discover new keywords in the process of your literature search.

Search for relevant sources

Use your keywords to begin searching for sources. Some useful databases to search for journals and articles include:

- Your university's library catalogue
- Google Scholar
- JSTOR
- EBSCO
- Project Muse (humanities and social sciences)
- Medline (life sciences and biomedicine)
- EconLit (economics)
- Inspec (physics, engineering and computer science)

You can also use boolean operators to help narrow down your search. Make sure to read the abstract to find out whether an article is relevant to your question. When you find a useful book or article, you can check the bibliography to find other relevant sources.

Step 2 – Evaluate and select sources

You likely won't be able to read absolutely everything that has been written on your topic, so it will be necessary to evaluate which sources are most relevant to your research question. For each publication, ask yourself:

- What question or problem is the author addressing?
- What are the key concepts and how are they defined?
- What are the key theories, models, and methods?
- Does the research use established frameworks or take an innovative approach?
- What are the results and conclusions of the study?
- How does the publication relate to other literature in the field? Does it confirm, add to, or challenge established knowledge?
- What are the strengths and weaknesses of the research?

Make sure the sources you use are credible, and make sure you read any landmark studies and major theories in your field of research.

Take notes and cite your sources

As you read, you should also begin the writing process. Take notes that you can later incorporate into the text of your literature review. It is important to keep track of your sources with citations to avoid plagiarism. It can be helpful to make an annotated bibliography, where you compile full citation information and write a paragraph of summary and analysis for each source. This helps you remember what you read and saves time later in the process.

Step 3 – Identify themes, debates, and gaps

To begin organizing your literature review's argument and structure, be sure you understand the connections and relationships between the sources you've read. Based on your reading and notes, you can look for:

- **Trends and patterns (in theory, method or results):** do certain approaches become more or less popular over time?
- **Themes:** what questions or concepts recur across the literature?
- **Debates, conflicts and contradictions:** where do sources disagree?
- **Pivotal publications:** are there any influential theories or studies that changed the direction of the field?
- **Gaps:** what is missing from the literature? Are there weaknesses that need to be addressed?

This step will help you work out the structure of your literature review and (if applicable) show how your own research will contribute to existing knowledge.

Step 4 Outline your literature review's structure

There are various approaches to organizing the body of a literature review. Depending on the length of your literature review, you can combine several of these strategies (for example, your overall structure might be thematic, but each theme is discussed chronologically).

Chronological

The simplest approach is to trace the development of the topic over time. However, if you choose this strategy, be careful to avoid simply listing and summarizing sources in order. Try to analyze patterns, turning points and key debates that have shaped the direction of the field. Give your interpretation of how and why certain developments occurred.

Thematic

If you have found some recurring central themes, you can organize your literature review into subsections that address different aspects of the topic. For example, if you are reviewing literature about inequalities in migrant health outcomes, key themes might include healthcare policy, language barriers, cultural attitudes, legal status, and economic access.

Methodological

If you draw your sources from different disciplines or fields that use a variety of research methods, you might want to compare the results and conclusions that emerge from different approaches. For example:

- Look at what results have emerged in qualitative versus quantitative research
- Discuss how the topic has been approached by empirical versus theoretical scholarship
- Divide the literature into sociological, historical, and cultural sources

Theoretical

A literature review is often the foundation for a theoretical framework. You can use it to discuss various theories, models, and definitions of key concepts. You might argue for the relevance of a specific theoretical approach, or combine various theoretical concepts to create a framework for your research.

Step 5 – Write your literature review

Like any other academic text, your literature review should have an introduction, a main body, and a conclusion. What you include in each depends on the objective of your literature review.

Introduction - The introduction should clearly establish the focus and purpose of the literature review.

Body - Depending on the length of your literature review, you might want to divide the body into subsections. You can use a subheading for each theme, time period, or methodological approach.

Conclusion - In the conclusion, you should summarize the key findings you have taken from the literature and emphasize their significance.

As you write, you can follow these tips:

- **Summarize and synthesize:** give an overview of the main points of each source and combine them into a coherent whole
- **Analyze and interpret:** don't just paraphrase other researchers—add your own interpretations where possible, discussing the significance of findings in relation to the literature as a whole
- **Critically evaluate:** mention the strengths and weaknesses of your sources
- **Write in well-structured paragraphs:** use transition words and topic sentences to draw connections, comparisons and contrasts.

When you've finished writing and revising your literature review, don't forget to proofread thoroughly before submitting.

Examples of literature reviews

Writing literature reviews can be quite challenging! A good starting point could be to look at some examples, depending on what kind of literature review you'd like to write.

- **Example literature review #1:** "Why Do People Migrate? A Review of the Theoretical Literature" (**Theoretical** literature review about the development of economic migration theory from the 1950s to today).
- **Example literature review #2:** "Literature review as a research methodology: An overview and guidelines" (**Methodological** literature review about interdisciplinary knowledge acquisition and production).
- **Example literature review #3:** "The Use of Technology in English Language Learning: A Literature Review" (**Thematic** literature review about the effects of technology on language acquisition).
- **Example literature review #4:** "Learners' Listening Comprehension Difficulties in English Language Learning: A Literature Review" (**Chronological** literature review about how the concept of listening skills has changed over time).

Analysis and Synthesis of Prior Art Bibliographic Databases

A good literature review doesn't just summarize sources it analyzes, synthesizes, and critically evaluates to give a clear picture of the state of knowledge on the subject.

Data analysis is the process of organizing and classifying the information you have collected, tabulating it, summarizing it, comparing the results with other appropriate information, and presenting the results in an easily understandable manner.

As you analyze your data, consider the following:

- What are the emerging patterns?
- Are there differences between certain groups or characteristics?
- Is the difference notable or statistically significant?
- Would the results be different if analyzed in another way (e.g., disaggregating, grouping)?
- What areas require additional analysis, information, or discussion with stakeholders?
- How will the data be displayed?

Below is an example of a data analysis plan that can help with planning

Variable	Data Transformation	Analytic Technique	Data Visualization
What is the variable being analyzed?	*What coding or transformation needs to happen?*	*What test, stratification, or other techniques will be used?*	*How will results be visualized?*

Data Synthesis brings together results and examines the findings together for patterns of agreement, convergence, divergence, or discrepancy. As part of this step, triangulating your findings involves organizing all of the results effectively. Finding the best way to organize, compare, and display all findings in a way that makes sense for your evaluation is an art and science. When triangulating, you are corroborating findings from different methods and perspectives, comparing the evidence, and dealing with discrepancies. Consider the following during data synthesis:

- Would the results be different if the evidence were collected differently (e.g., different data sources, groups, measures, design, time, level)?
- What areas require additional analysis, information, or discussion with stakeholders? • Are the findings consistent or contradictory?
- Are your results consistent with theories supported by previous evaluations or research?
- What are the limitations of your data analysis and interpretation process (e.g., potential biases, generalizability of results, reliability, validity)?
- If you used multiple indicators or different data collection methods, did you get similar results?

- How do your results compare with those of similar programs?

Synthesis Methods

Narrative summary: is a summary of the review results when meta-analysis is not possible. Narrative summaries describe the results of the review, but some can take a more interpretive approach in summarising the results. These are known as "evidence statements" and can include the results of quality appraisal and weighting processes and provide the ratings of the studies.

Meta-analysis: is a quantitative synthesis of the results from included studies using statistical analysis methods that are extensions to those used in primary studies. Meta-analysis can provide a more precise estimate of the outcomes by measuring and counting for uncertainty of outcomes from individual studies by means of statistical methods. However, it is not always feasible to conduct statistical analyses due to several reasons including inadequate data, heterogeneous data, poor quality of included studies and the level of complexity.

Qualitative Data Synthesis (QDS): is a method of identifying common themes across qualitative studies to create a great degree of conceptual development compared with narrative reviews. The key concepts are identified through a process that begins with interpretations of the primary findings reported to researchers which will then be interpreted to their views of the meaning in a second-order and finally interpreted by reviewers into explanations and generating hypotheses.

Mixed methods synthesis: is an advanced method of data synthesis developed by EPPI-Centre to better understand the meanings of quantitative studies by conducting a parallel review of user evaluations to traditional systematic reviews and combining the findings of the syntheses to identify and provide clear directions in practice.

Web of Science

The Web of Science is a bibliographic database of scholarly articles from 22,000 peer-reviewed journals worldwide. The platform provides tools for advanced search, citation analysis and bibliometrics. Access via this catalogue record. The Learning Resources' Hub provides self-paced training for the Web of Science. The Web of Science (WoS; previously known as Web of Knowledge) is a paid-access platform that provides (typically via the internet) access to multiple databases that provide reference and citation data from academic journals, conference proceedings, and other documents in various academic disciplines. Until 1997, it was originally produced by the Institute for Scientific Information. It is currently owned by Clarivate.

Web of Science is described as a unifying research tool that enables the user to acquire, analyze, and disseminate database information in a timely manner. This is accomplished because of the creation of a common vocabulary, called ontology, for varied search terms and varied data. Moreover, search terms generate related information across categories. Acceptable content for Web of Science is determined by an evaluation and selection process based on the following criteria: impact, influence, timeliness, peer review, and geographic representation. Web of Science employs various search and analysis capabilities:

- First, citation indexing is employed, which is enhanced by the capability to search for results across disciplines. The influence, impact, history, and methodology of an idea can be followed from its first instance, notice, or referral to the present day. This technology points to a deficiency with the keyword-only method of searching.
- Second, subtle trends and patterns relevant to the literature or research of interest, become apparent. Broad trends indicate significant topics of the day, as well as the history relevant to both the work at hand, and particular areas of study.
- Third, trends can be graphically represented.

Getting started with Web of Science

1. Access *Web of Science* from your library's electronic resources page.

2. Create a sign-in for *Web of Science* (top right corner of the landing page).

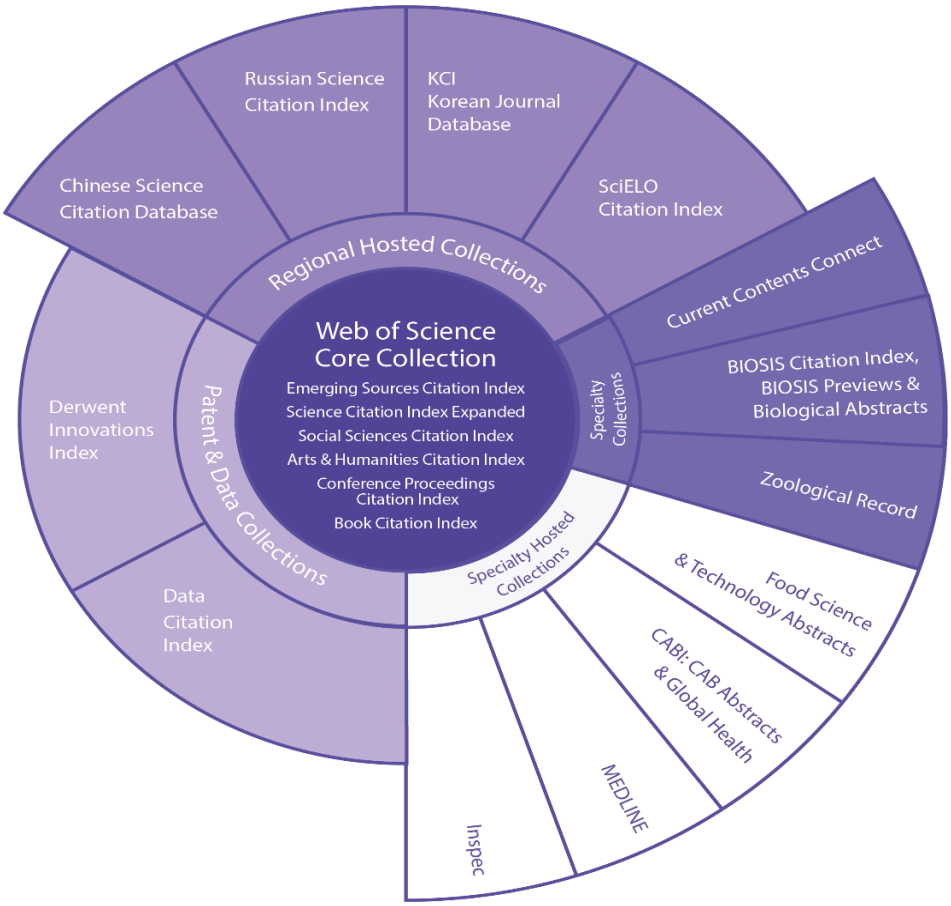
While optional, creating a sign-in allows you to:

- access *Web of Science* offsite (http://webofscience.com)
    - Roaming access is 6 months.
    - Sign in onsite to refresh roaming access for another 6 month period.
- save your searches
- set up search alerts and citation alerts
- save lists of records for later use
- sign in with the same credentials for **InCites**, **EndNote** online, and **Publons**

3. Sign up for live web-based training. Register for a class and walk through our products with an expert trainer.

4. Watch our video tutorials. This playlist of short videos from our YouTube channel walks you through key features on the Web of Science platform.

5. Use the tabs at the top of this guide to explore each database on the platform in more detail.



Google and Google Scholar

Google Scholar is a web search engine that specifically searches scholarly literature and academic resources. It provides a simple way to broadly search for scholarly literature. From one place, you can search across many disciplines and sources: articles, theses, books, abstracts and court opinions, from academic publishers, professional societies, online repositories, universities and other web sites. Google Scholar helps you find relevant work across the world of scholarly research.

How is google different from google scholar?

Google searches public Web content. "Don't use Google," means that you should not use the public Web content. Google Scholar is different. It searches the same kinds of scholarly books, articles, and documents that you search in the Library's catalogue and databases. The scholarly, authoritative focus of Google Scholar distinguishes it from ordinary Google.

So how is Google Scholar related to (and different from) the Library's databases?

There is overlap between the content in Google Scholar and the Library's individual databases. Also, many citations in Google Scholar will link to full text in the Library's databases or in publicly available databases. But Google Scholar will not contain everything that is in the Library's databases. Google Scholar can be a convenient starting place, but it is not a comprehensive "one-stop shop." For more precise searching, more search features, and more content, use the Library's individual databases.

How do I search and view items in Google Scholar?

Searching is as easy as searching in regular Google. Start from the Library's Homepage to search SHSU's Google Scholar. Click on the Articles & More tab and locate the Google Scholar search box at the very bottom. Enter a search term or phrase, such as "bird flu." Like regular Google, Google Scholar returns the most relevant results first, based on an item's full text, author, source, and the number of times it has been cited in other sources. Some actions are a little different from regular Google: clicking on a title may only take you to a citation or description, rather than to the full document itself. Google Scholar will not necessarily get you to the full text of every search result.

How do I find the full-text documents in my search results?

To find the full document, look for (1) a PDF or HTML link to the right of the article title, or (2) an Online Resources @ SHSU link. These links will help you find the full text of the document, either in a publicly available place or in one of the online databases offered by SHSU. If you don't see these links or they don't take you to the full text, you can contact the Library Service Desk for help in finding the article. Some documents will be unavailable online, but they may be available in the library building or through Interlibrary Loan.

Google Scholar: pros and cons

Google Scholar is a very powerful search engine for scientific literature that is used by many researchers and students. It is especially useful to find and access publications that you already know, or to do a quick search on a topic.

Google Scholar is less useful when you want to get an overview of literature on a certain topic, e.g., for your thesis or literature review. This is because Google Scholar offers limited options to combine multiple search terms with Boolean operators (like AND, OR, NOT). By default, Google Scholar searches in the full text of publications. Advanced searching allows you to limit your search to specific fields (title, author, a particular journal and date), but you can't limit your search to e.g. title, abstract and keywords fields only (as in Scopus).

The selection that Google Scholar makes for you is not transparent. It ranks the search results and shows only the first 1,000 results of any search, based on algorithms that Google changes frequently. The ranking depends on settings that you may be unaware of, such as your language settings or location. To get an overview of scientific literature on a certain topic, therefore, it's better to use a bibliographic database. Find out here how to choose the best databases for your subject.

Getting access to publications

If you use Google Scholar from the WUR Library website, you automatically get access to the sources that are part of the WUR collections, if you are on campus. When you work from the Google Scholar website, make sure that it makes a link to WUR Library to give you easy access to the licensed sources. If you don't see a 'Get It from WUR' link next to your search results, go to Settings in the menu on the top of the page. Here, choose Library links and add Wageningen University & Research Library to the list. For more information about access to licensed sources off-campus, go to How to access licensed sources directly.

Tips to improve your searching

Note that most of these tips also work in Google!

- Use the **Advanced search** option (in the menu) to search in specific 'fields' or to limit results by year range. These options won't work optimally (see above), but it can help to limit the number of results.
- Use double quotation marks to search for **multiple words** next to each other in the specified order (like in compound terms or an exact phrase), e.g., "climate change" or "the impact of climate change on food security". Otherwise, Google (Scholar) automatically combines multiple words with the operator AND.
- **Include alternative terms** by using the OR operator. In some cases Google (Scholar) doesn't include obvious synonyms in your search. With the OR operator you can combine these terms and find more. Instead of OR you can also use | (a pipe), e.g., "heart|myocardial infarction|attack" finds heart infarction, myocardial infarction, heart attack and myocardial attack.
- **Exclude specific terms** by using the – operator. You can exclude as many terms as you want, e.g., mercury –ford –freddy –outboards –planet.
- **Allintitle**: Limit your search to terms appearing in the title only, e.g., allintitle:"agaricus bisporus".
- **Filetype**: Limit your search to specific file types by using filetype: or ext: E.g., "agaricus bisporus" filetype:pdf
- **Site**: Limit your search to certain websites or domains. This can be useful for websites without good search options, e.g., "plant diseases" site:journals.plos.org. By searching within certain domain extensions, you can limit your search by country or type of institution, e.g., "plant diseases" site:.edu (academic institutions in the USA).
- **Combine** all of the above to do more precise searches, e.g., allintitle:"carbon dioxide" OR CO2 -phosphorus ext:pdf site:.edu
- **Personalise** your searching via Settings and use other handy features of Google Scholar. For example, make your own library of references (called My Library), create literature alerts, or let Google Scholar show import citation links to EndNote or another reference manager. To use these options, you'll need to sign in with your Google account.

Features of Google Scholar

- Search all scholarly literature from one convenient place
- Explore related works, citations, authors, and publications
- Locate the complete document through your library or on the web
- Keep up with recent developments in any area of research
- Check who's citing your publications, create a public author profile

Effective Search: The Way Forward Introduction to Technical Reading Conceptualizing Research

An effective search is one that:

- Returns results on highly likely matches or on the correct individuals
- Doesn't overwhelm you with too many results or underwhelm you with too little or no results.

The desire to undertake effective searches has strong links to the Genealogical Proof Standard and its goal of performing a 'reasonably exhaustive' search.

This means that you have examined a wide range of high-quality sources in an organised way, considering name variations and possible confusion with other likely matches.

A searching strategy

A good way is to make sure that you have done an effective search is to formulate a searching strategy before you begin searching and to keep refining it as you go along.

A searching strategy has several steps:

Step #1 – Analyse your topic to decide where to begin - Does the topic have distinctive words? E.g.: Hesse-Cassel, Testatika – Have variant spellings, synonyms, etc? E.g.: Bolander, Bolender, Bolland or Millennium, Year 2000.

Step #2 – Choose the right starting place…what databases or other search tools do you want to use?

Step #3 – Learn as you go & VARY your approach with what you learn.

Step #4 – Don't get stuck on a strategy that doesn't work – Be willing to start again and abandon fruitless searches. Switch from one search tool to another.

Step #5 – Return to the search better informed.

(Many thanks to the U.C. Berkeley Library for these strategies)

Some quick tips on effectively searching genealogy databases

- Search data collections separately on large multi-collection database sites such as Ancestry or FamilySearch. For example, just search the 1891 Canadian census collection or the Australian electoral roll collection search pages rather than starting a search with the general search boxes on the front page.
- Don't enter everything you know about a person right off; just start with their name and perhaps a place of birth and range of birth dates. You can always enter more data later if you get loads of possible matches
- Don't get side-tracked from your primary research goal. Note down interesting bits of information and go back later to check them out.
- Use Boolean searching techniques if the database allows this. Boolean terms are words used to connect terms you want to search with. They are: AND, OR and NOT. –AND requires all terms to appear in a record. Using AND narrows a search and thus returns fewer results. Many databases use AND as their default. –OR retrieves records that include either term searched for. Using OR broadens a search and thus returns more results. –NOT excludes terms found within records. Using NOT narrows a search and thus returns fewer results.
- Remember to try using specialist databases rather than just sticking to the same old ones you normally use. We'll cover some of these in the course but Cyndi's List is a great way to discover new sources of information. And do remember to search for people in regular internet search engines such as Google or DuckDuckGo; it's amazing what can be found out by doing this.

Critical and Creative Reading

Critical reading - Critical reading means that a reader applies certain processes, models, questions, and theories that result in enhanced clarity and comprehension. There is more involved, both in effort and understanding, in a critical reading than in a mere "skimming" of the text. What is the difference? If a reader

"skims" the text, superficial characteristics and information are as far as the reader goes. A critical reading gets at "deep structure" (if there is such a thing apart from the superficial text!), that is, logical consistency, tone, organization, and a number of other very important sounding terms.

What does it take to be a critical reader? There are a variety of answers available to this question; here are some suggested steps:

1. Prepare to become part of the writer's audience.

After all, authors design texts for specific audiences, and becoming a member of the target audience makes it easier to get at the author's purpose. Learn about the author, the history of the author and the text, the author's anticipated audience; read introductions and notes.

2. Prepare to read with an open mind.

Critical readers seek knowledge; they do not "rewrite" a work to suit their own personalities. Your task as an enlightened critical reader is to read what is on the page, giving the writer a fair chance to develop ideas and allowing yourself to reflect thoughtfully, objectively, on the text.

3. Consider the title.

This may seem obvious, but the title may provide clues to the writer's attitude, goals, personal viewpoint, or approach.

4. Read slowly.

Again, this appears obvious, but it is a factor in a "close reading." By slowing down, you will make more connections within the text.

5. Use the dictionary and other appropriate reference works.

If there is a word in the text that is not clear or difficult to define in context: look it up. Every word is important, and if part of the text is thick with technical terms, it is doubly important to know how the author is using them.

6. Make notes.

Jot down marginal notes, underline and highlight, write down ideas in a notebook, do whatever works for your own personal taste. Note for yourself the main ideas, the thesis, the author's main points to support the theory. Writing while reading aids your memory in many ways, especially by making a link that is unclear in the text concrete in your own writing.

7. Keep a reading journal

In addition to note-taking, it is often helpful to regularly record your responses and thoughts in a more permanent place that is yours to consult. By developing a habit of reading and writing in conjunction, both skills will improve.

Critical reading involves using logical and rhetorical skills. Identifying the author's thesis is a good place to start, but to grasp how the author intends to support it is a difficult task. More often than not an author will make a claim (most commonly in the form of the thesis) and support it in the body of the text. The support for the author's claim is in the evidence provided to suggest that the author's intended argument is sound, or reasonably acceptable. What ties these two together is a series of logical links that convinces the reader of the coherence of the author's argument: this is the warrant. If the author's premise is not supportable, a critical reading will uncover the lapses in the text that show it to be unsound.

Creative Reading:

The process of reading beyond the text to infer and react to what is read, providing fresh and original thoughts not clearly stated in the material, is known as creative reading. It requires creativity, ingenuity, and originality. Creative reading differs from critical reading in that it requires creative thinking abilities. Creative reading is a reading technique that stresses active involvement and interpretation of a text rather than passively absorbing its content. It entails connecting with the material on a deeper level and exploring its themes, ideas, and implications utilizing imagination and critical thinking skills.

Visualizing scenes and characters, creating connections to personal experiences, evaluating language and literary methods, and asking questions about the text are all examples of creative reading techniques. It

may also entail investigating various interpretations and views, as well as taking into account the social, historical, and cultural context in which the book was created.

Creative reading is not confined to works of fiction; it can also be applied to nonfiction, poetry, and academic writing. Reading is an useful ability that can improve both the enjoyment and comprehension of reading, as well as the development of critical thinking, empathy, and creativity.

What is the significance of creative reading?

Creative reading is beneficial because it helps readers to use their imagination to create new meanings and concepts. It entails reading for suggested and inferred meanings, appreciative reactions, and critical evaluation, all of which assist the reader in become an active participant in the process of developing new ideas. Reading can also help you focus and be more creative.

How to improve your creative reading abilities?

Here are some pointers on how to improve your creative reading abilities:

1. Visualize scenes and characters: As you read, try to visualize the scenes and characters in your mind. This can help bring the text to life and make it more engaging.
2. Make connections to personal experiences: Look for ways to connect the text to your own experiences or emotions. This can help you relate to the characters and themes on a more personal level.
3. Analyze language and literary devices: Pay attention to the language and literary devices used in the text, such as metaphors, imagery, and symbolism. Try to think about why the author used these devices and what they might be trying to convey.
4. Ask questions: As you read, ask yourself questions about the text. What do you think will happen next? Why did the character make that decision? What is the author trying to say?
5. Consider different interpretations and perspectives: Try to think about the text from different perspectives and consider different interpretations. This can help you develop critical thinking skills and a deeper understanding of the text.

6. Experiment with different reading techniques: Try experimenting with different reading techniques, such as speed reading or reading out loud. This can help you engage with the text in different ways and find the approach that works best for you.
7. Practice regularly: Like any skill, creative reading takes practice. Make a habit of reading regularly and try to incorporate some of these techniques into your reading routine. Over time, you'll find that your creative reading skills improve and your enjoyment of reading deepens.

It is critical to actively and imaginatively engage with the material in order to build creative reading skills. Visualizing scenes and people, finding personal connections, evaluating language and literary methods, asking questions, considering multiple interpretations and perspectives, experimenting with different reading techniques, and practicing regularly can all help. You may increase your comprehension and pleasure of literature, build critical thinking abilities, and cultivate creativity and empathy by adding these tactics into your reading practice.

Taking Notes While Reading

The importance of good notes

Taking good notes while reading is an important part of academic success in college. Most courses require significant reading, and it can be difficult to understand and master the material and do well in class without solid note taking and reading skills.

Good notes from your reading can help you:

- organize your ideas and information from the text
- keep focused and stay engaged while reading
- keep a record of what you read so you can more easily locate it in the future
- think critically about what you read while you read
- draw conclusions and identify main ideas of the text
- be prepared for class and build a foundation for lecture
- have solid materials to use to study for exams or prepare for assignments

How to take good notes while reading

Good notes can take different forms and may vary from person to person or even from text to text. One of the key ideas of good note taking is that it is not necessary to copy down loads of information from the text. Copying down information does not engage your brain and is not a strong strategy for learning and remembering content. It also takes a lot of time and energy. In contrast, simply highlighting loads of information is simpler but does not do much to actively engage the brain. Instead of copying down tons of notes or over-highlighting, try some of the active and effective strategies and formats listed below. These will help you decrease the amount of time and energy you spend on notes and increase your comprehension and retention from reading.

Different formats/strategies for notes

There is no one right way to take notes while reading. The important thing is that you experiment with a few effective strategies, find some that work for you, and use them. You may find that different formats or strategies work better for different types of texts, too, and you may want to use different ones for different classes. Below are some examples to try:

Try taking notes from memory

- Students often miss the opportunity to digest the information from their texts because they're too busy worrying about taking good notes—instead of actually comprehending the content, they're thinking more about what they should write down.
- Try reading short sections of your reading (likely a paragraph or two or up to a page) and pausing to think about what you just read—then take notes from your memory of what you just read. This will help you focus on the main points instead of getting caught up in details.
- If you are taking notes digitally or online, try creating a Word or Google Document for each article or textbook chapter you have read. Write a summary of the key points from each reading without looking back at the text. Then fill in the details by consulting the reading only after you first have recalled everything you can.
- It's okay to not remember 100% of what you just read; focus on the main points, and then refer back to the text to fill in details as needed.

- This method may take slightly longer, but many students say it's worth it due to the increase in reading comprehension.

Mark directly on the text

- If you have a print version that allows it, simply use a pen or pencil. For online texts, some digital programs also allow annotating, highlighting, and commenting.
- Underline, circle, or highlight key words and phrases this can be helpful for students who need to do something with their hands to help them stay focused.
- Annotate margins with symbols, abbreviations, or summaries of the text in your own words. See our annotating handout for more explanation.
- If you have an online text, you can still record your thoughts, key words, and summaries in this way. Just grab a plain sheet of paper, label it with the text and chapter/page number, and jot them down on the paper instead of in the book.

Cornell style notes

- Divide a piece of paper into three sections approximately two inches blank at the bottom, and the top portion divided into a one-third section on the left and a two-third section on the right.
- Take notes on the right two-thirds of the page.
- List key words or questions in the left column.
- Summarize the entire page in the space at the bottom.
- Learn more by reading a more detailed explanation of Cornell style notes, or try out of these MS Word templates.

App-based notes (EverNote and OneNote)

- EverNote is a popular notetaking app that can sync notes on all your devices. You can organize notes into individual "notebooks" relevant to different classes or topics. Its fans find it useful for general note-taking and class notes as well as for daily tasks and organization.

- OneNote is an app that comes with Microsoft Office, which is free for UNC students! If you have not downloaded it yet, you can download OneNote here. Much like EverNote, OneNote allows you to sync your notes and access them on any device. You can annotate notes, clip and save multimedia content, and record audio notes.

Create a graphic organizer or concept map

- This method is good for texts that have a lot of higher level concepts that require explanations or texts that have remember-level facts, dates, terms, etc.
- Organize information visually.
- Differentiate main ideas from support in an appropriate format: concept map, table, flow chart, hierarchy, timeline, or Venn diagram.
- Good for texts that have a lot of visuals, timelines, etc. like science or history.
- Generate Your Own Q&A or Study Guide.
- Formulate questions from headings and keywords before you begin. Then seek answers as you read
- Are you studying for test a that will require a lot of visual identification? Consider creating your own Pinterest board to organize image-based notes.

Helpful tips

- Write notes in your own words instead of copying down information from the book.
- Avoid over-highlighting. Highlighting doesn't do much to actively engage the brain, so it's not the most useful strategy. Also, highlighting too much can keep you from focusing on the main ideas. For tips on highlighting more thoughtfully, see the strategies on our highlighting handout.
- Wait until the end of a page to take notes so that you can better focus on what you are reading and so that you can try to summarize in your own words rather than copy.
- You don't need to write pages of notes keep them brief and focused.
- Preview the chapter before you start reading by looking at the text features to gain clues about the main ideas of the chapter.
- Focus on the main ideas and concepts.

Reading mathematics and algorithms:

Mathematical Algorithm can be defined as an algorithm or procedure which is utilized to solve a mathematical problem, or mathematical problem which can be solved using Digital Signature Algorithm. In the field of DSA, mathematical algorithms refer to those algorithms which are generally used to solve problems related to the field of computer science.

What are some use cases of Mathematical Algorithms?

It is a very useful topic for solving problems of computer science and also has vast applications in competitive programming. Some of them are mentioned below:

1. Problem-Solving: It is mostly used in solving complex mathematical ideas in mathematics and computer science.

2. Competitive Programming: Algorithms like Sieve of Eratosthenes, Euclid algorithm, Binary exponentiation are some of the most common topics used in competitive programming.

3. Recurrence relation: A lot of problems in DSA require the usage of recursion. Using the knowledge of mathematics several recurrence relations can be turned into mathematical formulae which makes solving the problem quite an easy task.

4. Heavy calculation: Problems containing complex mathematical concepts and heavy calculations are easily done in comparatively less time using these algorithms instead of manual calculations.

5. Statistics: Mathematical algorithms are also important for data processing, i.e., for converting raw data into useful information and also for analyzing them.

Reading a Datasheet:

Datasheets are instruction manuals for electronic components. They (hopefully) explain exactly what a component does and how to use it. Unfortunately, these documents are usually written by engineers for other engineers, and as such they can often be difficult to read, especially for newcomers. Nevertheless, datasheets are still the best place to find the details you need to design a circuit or get one working. A datasheet's contents will vary widely depending on the type of part, but they will usually have most of the following sections:

- The first page is usually a **summary** of the part's function and features. This is where you can quickly find a description of the part's functionality, the basic **specifications** (numbers that describe what a part needs and can do), and sometimes a **functional block diagram** that shows the internal functions of the part. This page will often give you a good first impression as to whether potential part will work for your project or not.
- A **pinout** lists the part's pins, their functions, and where they're physically located on the part for various packages the part might be available in. Note the special marks on the part for determining where pin 1 is (this is important when you plug the part into your circuit!), and how the pins are numbered (the below parts are numbered counterclockwise). You'll find some acronyms here: VCC is the supply voltage (commonly 5V or 3.3V), CLK is clock, CLR is clear, OE is output enable, etc. These acronyms should be spelled out later in the datasheet, but if not, try Google or Wikipedia. If a pin has a star next to it or a line over the name, that's an indication that the pin is *active low* which means that you'll pull the pin low (0V) to activate it, rather than H (VCC).
- Detailed tables of electrical specifications follow. These will often list the **absolute maximum ratings** a part can withstand before being damaged. Never exceed these or you'll be replacing a possibly expensive part!
- You'll also see the more normal **recommended operating conditions**. These may include voltage and current ranges for various functions, timing information, temperature ranges, bus addresses, and other useful performance information. The below excerpt contains a good example where the fine print can help you out: "Note 3" in this set of specifications states that "All unused inputs of the device must be held at VCC or GND to ensure proper device operation." This is a reminder to tie all unused inputs H or L to prevent them from "floating" between H and L which can make your circuit malfunction and be difficult to debug.
- Some parts will have one or more **graphs** showing the part's performance vs. various criteria (supply voltage, temperature, etc.) Keep an eye out for "safe zones" where reliable operation is guaranteed.

- **Truth tables** show how changing the *inputs* to a part will affect its *output*. Each line has all the part's inputs set to specific states, and the resulting output of the part. "H" means that input is a logical high (usually VCC), "L" means a logical low (usually GND), "X" means the chip doesn't care what the input is (could be H or L), and an arrow means that that you should change the state of that pin from L to H or H to L depending on the arrow direction. This is called "clocking" an input, and many chips rely on this for proper operation.
- **Timing diagrams** show how data should be sent to and received from the part, and what speed it should be sent / received. These are typically laid out with various inputs and outputs as horizontal lines, showing the logic transitions that happen to those lines over time. If the trace dips down, that's a L input or output. If the line rises higher, that's a H input our output. Timing specifications are laid out as arrows between transitions (names are referenced back to timing numbers in the electrical specs), and vertical bars or arrows will link related transitions.
- Complex parts will have extensive **application information**. This varies depending on the part, but may include detailed descriptions of pin functions, how to communicate with the part, lists of commands, memory tables, etc. This is often very useful information, so read through it carefully.
- Some datasheets will include **example schematics** for various circuits that can be built around the part. These are often very useful building blocks for interesting projects, so be sure to look through them.
- Some parts are sensitive to the way they're built into a circuit, and the datasheet will provide **layout considerations**. These can range from noise-reduction techniques, to dealing with thermal issues, to mechanical mounting considerations as with the accelerometer below. This all tends to be very good advice, that if followed from the start will lead to the most trouble-free circuits. Likewise, if you don't follow this advice, your circuit may have problems later on that can be hard to diagnose, and harder to fix.
- At the end of many datasheets is **packaging information,** which provides accurate dimensions of the packages a part is available in.

- Finally, a few of our customers have correctly pointed out that datasheets are subject to having errors just like anything else, and running into one of these errors can be frustrating to say the least. To reduce this possibility, be sure you have the latest version of a datasheet before doing any serious work. These are available at the manufacturer's website. And if nothing else helps, many manufacturers have applications engineers you can contact to get help on hard-to-solve problems.

Attributions and Citations

Giving Credit Wherever Due

If you say 'credit where credit's due', you are admitting that you ought to praise someone for something that they have done or for a good quality that they possess.

Give credit where credit is due - It sounds obvious but it doesn't always happen. Especially in highly political and hierarchical organizations, people often get credit based on their power, not their actual contributions. Look closely at your team or organization to identify the biases that cause some people's work or ideas to be overvalued or undervalued. Then make sure that the right people are getting the right credit for the right reasons at the right time.

Citations: Functions and Attributes

What is a citation?

Citations are a way of giving credit when certain material in your work came from another source. It also gives your readers the information necessary to find that source again. It provides an important roadmap to your research process. Whenever you use sources such as books, journals or websites in your research, you must give credit to the original author by citing the source.

Why do researchers cite?

Scholarship is a conversation and scholars use citations not only to give credit to original creators and thinkers, but also to add strength and authority to their own work. By citing their sources, scholars are placing their work in a specific context to show where they "fit" within the larger conversation. Citations are also a

great way to leave a trail intended to help others who may want to explore the conversation or use the sources in their own work.

In short, citations

(1) give credit

(2) add strength and authority to your work

(3) place your work in a specific context

(4) leave a trail for other scholars

"Good citations should reveal your sources, not conceal them. They should honeslty reflect the research you conducted."

Impact of Title and Keywords on Citations

Citation performance of a publication depends heavily on its academic field. Some words in keywords, titles, and abstracts of publications may be indicative of their academic field. Therefore, analysis of differences in citation performance of these words helps us understand inter-field differences in citation performance.

Citation performances of words were computed separately for keywords, titles, and abstracts. Therefore, a specific word such as “data” may have a different performance depending on whether it comes from keywords, titles, and abstracts. 2010 publications have more time to attract citations than 2012 publications. Because of this, citation performance of a publication is normalized by its publication year. The following formula was used to compute normalized citation performance of publications by their keywords.

a) “w” is the keyword. “y” is the publication year.

b) C_y^w is the total citation of publications that contain keyword w in the year y.

c) N_y^w is the total number of publications that contain keyword w in the year y.

d) C_y is the total number of citations that are received by all keywords in the year y. If there are S separate keywords in a publication, then citation of that publication is counted S times.

e) N_y is the total number of all keywords in the year y. If there are S separate keywords in a publication, then that publication is counted S times. Therefore, Cy/N_y is the publication performance of an average keyword in the year y.

f) A_y^w is the average publication performance of w in the year y. It is computed as follows:

$$A_y^w = \frac{C_y^w / N_y^w}{C_y / N_y}$$

If A_y^w is greater than 1, then w has a higher average citation than average keyword of that year.

g) Year-adjusted keyword performance (YK) is the average performance of w through years. It is simply computed by taking the averages of all years that w is used. If w is used in all 3 years, then the formula of YK is as follows:

$$YK = \frac{\sum_y A_y^w}{3}$$

If we use the keyword "data"as w. Then, YK measures the average citation performance of all publications that have keyword "data" relative to the average citation performance of all keywords. If YK is above 1 for a certain keyword, we can say that the citation performance of that keyword is above the average citation performance of all keywords.

A simple variation of the abovementioned formula was used to compute performances of publications by their titles and abstracts. Instead of using a certain keyword for "w" in the abovementioned formula, we used a certain word in the titles or abstracts of the publication. If "w" stands for a certain word in the title, then the resulting average performance is called "year-adjusted title word performance" (YT). Alternatively, if "w"stands for a certain word in the abstract, then the resulting average performance is called "year-adjusted abstract word performance" (YA).

YK, YT, and YA can have different values even for the same word. If we choose w to be "empirical," then, YT gives the average performance of publications that contain "empirical" in their titles. This value can be different than YA which gives the average performance of publications that have "empirical" in their abstracts.

Knowledge flow through Citation

Knowledge flow between scientific disciplines has commonly been measured based on citation data. Previous studies using citing relationships have mostly considered direct citations but have paid little attention to indirect citations (IDC) to indicate how knowledge diffusion from one discipline to another via one or more

intermediaries. In this study, we measured knowledge flow between disciplines from two perspectives: direct citations (DC) and discipline potential energy (DPE), which is proposed to combine both direct and indirect citations. Data were collected from the Web of Science (WoS) database.

Example findings include: (1) DPE overshadows previous measures by considering not only direct citations but also indirect citations between disciplines which was usually ignored in previous measures, and revealed that the knowledge contribution of some disciplines had been underestimated by previous measures, such as Physics and Engineering. (2) The proportion of IDC contribution is close to that of direct knowledge contribution when the discipline scale is removed, which suggests that it is essential to consider IDC to distinguish the knowledge relationship (net-outflow/inflow) between disciplines. (3) Both measurements show that Biology & Biochemistry has always been the top discipline with the highest net outflow of knowledge, which is inconsistent with the history of science that Mathematics, Physics and Chemistry would be the highest net outflow disciplines. The results show that even considering IDC does not fully reveal the knowledge contribution and academic influence of disciplines. Therefore, caution should be taken in the use of citations as a primary measure of knowledge flow.

Citing Datasheets

The citation style for citing datasheet is given by: Company Name, "Title of data sheet," Data Sheet number, Date of Publication.

Material Safety Data Sheets

Title of MSDS/SDS; Number of MSDS/SDS [Print or Online]; Name of Manufacturing Company: Location of Company, Date of Publication, URL (date accessed).

E.g. Texas Instruments, "High speed CMOS logic analog multiplexers/demultiplexers," 74HC4051 datasheet, Nov. 1997 [Revised Sept. 2002].

E.g. *Ammonium Chloride*; SDS No. 000001616 [Online]; Avantor: Center Valley, PA, Feb 23, 2015. https://www.avantorinc.com/Documents/MSDS/USA/SAP/00038387.PDF (accessed April 27th, 2018).

Styles for Citations

Different subject disciplines call for citation information to be written in very specific order, capitalization, and punctuation. There are therefore many different style formats.

1. American Psychological Association (APA) Style

This style guide is generally used by individuals associated with behavioral and social sciences. It is a complete set of rules and regulations, along with guidelines to various informative elements. It is a complete set of a citation style guide that has detailed information about every step in the writing process.

E.g., Whisenant, W. A. (2003) How Women Have Fared as Interscholastic Athletic Administrators Since the Passage of Title IX. *Sex Roles*, 49 (3), 179-182.

2. American Medical Association (AMA) Style Guide

American Medical Association (AMA) style is used in the medical industry. You may refer to it for writing and editing medical research papers. The AMA Manual of Style was first published in 1963 and is currently in its 11th edition. AMA Style is a variation of the Vancouver system used by the Journal of the American Medical Association (JAMA) and some other publications by the AMA. AMA adopts a documentary-note style, which means you should put a number in your text to cite a source, and the reference list should be in numerical order.

3. Council of Science Editor Documentation Style (CSE)

This style guide is for the scientific industries. You may refer to it for editing and proofreading content related to chemistry, physics, biological sciences, and astronomy. Naturally, CSE's main point of focus is numbers, statistics, units, and mathematical expressions. In its recent edition, it includes information on the management of indexes, tables, figures, and dealing with copyright laws.

4. The Bluebook

The Bluebook is the style guide for the law industry in the United States used by law students, prosecutors, and judges use. The Bluebook provides systematized guidelines on how to structure and form written content, and it is the guidelines governing how the US legal documents are cited in legal memoranda, court documents, and law journals. It is published by the editors of the law journals at Columbia University, Harvard University, the University of Pennsylvania, and Yale University.

5. The Canadian Style

The Canadian Style is created by the Canadian government for the Canadian public. According to Public Services and Procurement Canada, *The Canadian Style* gives concise answers to questions concerning written English in the Canadian context. It covers such topics as the decimal point, abbreviations, capital letters, punctuation marks, hyphenation, spelling, frequently misused or confused words, and Canadian geographical names. It also includes useful advice for drafting letters, memos, reports, indexes, and bibliographies. In addition, *The Canadian Style* includes techniques for writing clearly and concisely, editing documents, and avoiding stereotyping in communications.

Also, due to its concise nature, it is actively used for editing and proofreading government content in English. It focuses on various punctuation marks, spelling, and misused words, along with providing detailed guidelines for writing various official documents. It follows the rules to give a clear, well-communicated writing style that the public can easily follow.

6. The Chicago Manual of Style

This style guide is perhaps one of the most famous citation styles in the academic field. Nicknamed as the ‘‘Editor’s Bible,’’ or ‘‘Turabian Style,’’ the Chicago Manual of Style is the most used in the publishing industry. The Chicago Style Manual was originally created by the University of Chicago and published in 1906. It was essentially written as a reference for authors, editors, copywriters, publishers, and other areas where the written word is primarily used. It is also used by other disciplines.

There are two Chicago Manual Styles: Notes-Bibliography System and the Author-Date System. Notes-Bibliography System is commonly used in literature, history, and the arts, while the Author- Date System is often used in the physical, natural, and social sciences.

E.g., Whisenant, Warren A. "How Women Have Fared as Interscholastic Athletic Administrators Since the Passage of Title IX." *Sex Roles* 49, no. 3 (2003): 179-182.

7. Modern Language Association (MLA) Style

Modern Language Association (MLA) is considered one of the most popular and most used citations and referencing styles in the academic field. MLA style is a style used for citing sources within the language

arts, cultural studies, and other humanities disciplines in the academic field. The latest version of the MLA manual (9th edition) offers several examples for the general format of MLA papers,in-text citations, and the Works Cited page. This quick guide provides the basic rules for MLA in-text citations for academic papers and essays.

E.g., Whisenant, Warren A. "How Women Have Fared as Interscholastic Athletic Administrators Since the Passage of Title IX." *Sex Roles* Vol. 49.3 (2003): 179-182.

8. Associated Press (AP) Stylebook

Similar to the Chicago Manual of Style, the Associated Press Stylebook is known as the ‘‘Journalists’ Bible.’’ Though the publication industry, editors, and even students use it, it is primarily for journalists. It provides fundamentals adhering to the journalist’s style. The 55th edition of The Associated Press Stylebook and Briefing on Media Law was published in 2020. According to AP official website, it includes more than 200 new or revised entries, with chapters covering data journalism, business, religion, and sports terms, as well as media law, news values, punctuation, social media, and polls and surveys, plus a new chapter on digital security for journalists.

Acknowledgements and Attributions

As nouns the difference between acknowledgment and attribution is that acknowledgement is the act of acknowledging; admission; avowal; owning; confession while attribution is the act of attributing something.

Acknowledgements	Attributions
• The act of acknowledging; admission; avowal; owning; confession. • The act of owning or recognized in a particular character or relationship; recognition as regards the existence, authority, truth, or genuineness. • The owning of a benefit received; courteous recognition; expression of thanks.	• The act of attributing or ascribing, as a quality, character, or function, to a thing or person, an effect to a cause. • That which is ascribed or attributed.

• Something given or done in return for a favor, message, etc. • A declaration or avowal of one's own act, to give it legal validity; as, the acknowledgment of a deed before a proper officer. Also, the certificate of the officer attesting such declaration.	

What should be Acknowledged?

The Acknowledgments section is present in both a paper and an academic thesis. For papers, the Acknowledgments section is usually presented at the back, whereas in a thesis, this section is located towards the front of the manuscript and is commonly placed somewhere between the abstract and Introduction. However, the exact location varies between each university, as each establishment possesses its own style guide for theses and student submissions. So, it is always worthwhile consulting your university's academic style guide before writing a manuscript for undergraduate/postgraduate submission.

Whom to Acknowledge?

1. Other authors/contributors: It is not common practice for the lead paper writer (i.e. the person writing and publishing the manuscript) to acknowledge the other authors/direct contributors to the paper. Only those who are not recognized as authors may be thanked and acknowledged.

- **Reviewers**: Authors are also not allowed to thank reviewers personally, or those who inspire them but cannot directly receive their appreciation – although reviewers can be thanked if they are kept anonymous.

Acknowledgement in Books, Dissertations

For academic thesis and Books, there is no right or wrong way to acknowledge people, and who you want to acknowledge is down to personal preference. However, the common types of people authors acknowledge in their academic thesis include:

- Supervisor's contributions
- Research group (especially if the thesis in question is a master's and the work is helped along by a PhD student)

- Support staff (laboratory technicians, etc.)
- Any students who undertook side projects with them (e.g. final year undergraduates, summer students, master's students)
- Administrative staff (there can be a lot of bureaucracy for thesis submissions)
- Referees that got them onto the course (postgraduate only)
- Funding bodies
- Any collaboration with industry and the people they worked with at said establishment(s)
- Friends
- Colleagues
- Family

Dedication or Acknowledgements

A *dedication* is a message from the author prefixed to a work in tribute to a person, group, or cause. Most dedications are short statements of tribute beginning with "To…" such as "To my family".

Acknowledgements are the author's statement of gratitude to and recognition of the people and institutions that helped the author's research and writing.

Any of the pages must be prepared following these guidelines:

1. Do not place a heading on the dedication page.
2. The text of short dedications must be centered and begin 2″ from the top of the page.

3. Headings are required for the "ACKNOWLEDGEMENTS" page. Headings must be in all capital letters and centered 2″ below the top of the page.
4. The text of the acknowledgements page must begin one double-spaced line below the heading, be double-spaced, and be aligned with the document's left margin with the exception of indenting new paragraphs.
5. Subsequent pages of text return to the 1″ top margin.
6. The page(s) must be numbered with consecutive lower case Roman numerals (starting with the page number after the abstract) centered with a 1/2″ margin from the bottom edge.

E.g, I dedicated my PhD thesis to several members of my family, and wrote short personal notes about their contributions. In my acknowledgements section, I thanked my supervisor, a mentor, a friend who provided technical advice during my research, and my scholarship provider.

. . . . . END . . . . . .